

SPECIFICATIONS

END OF STUDIES (PFE) PROJECT

Programme Grande Ecole

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CHAPTER I – GENERAL PRESENTATION

The final project is prepared simultaneously with the company mission, which normally constitutes its essential basis. It represents the culmination of the Programme Grande Ecole (PGE) and, as such, contributes 12 ECTS credits towards obtaining the diploma.

Below are some elements of context and objectives associated with the final studies project.

- General objectives

- Demonstrate mastery of the skills acquired during the PGE course
- Solve a concrete problem related to data management and analysis
- Integrate theoretical and practical approaches to AI and data science techniques

- Choice of subject

- Must be directly linked to the fields of Artificial intelligence/data science
- Prioritize innovative subjects with high added value
- Possibility of collaborating with the work-study company

The subject of the PFE must result above all from the student's readings, meetings, practices, tastes and reflections. The process of choosing the subject can also result from the work to be carried out in the company, the opinion of the learning tutor and the student's career aspirations.

It would be interesting to have a good harmony between the work carried out in the company and the subject of the PFE. The greater the synergies, the easier the work is, well supported by the company because more expected and therefore helped. The apprentice has every interest in playing the search for this synergy.

CHAPTER II – Writing and structure of the thesis

Although the form may vary depending on the issues and possibly the practices observed for the empirical dimension in the matter, it is nevertheless imperative that the thesis includes:

1 - A problem

This part corresponds to a description of the questions asked and the issues identified, specifically related to the management, exploitation, and analysis of data. This may include, for example:

- Of a hypothesis to be validated concerning the performances or biases of an analytical or predictive model.
- From a questioning or optimization of current data processing methods in a professional context.
- From an exploration of the limits or opportunities offered by new technologies or approaches in data analysis.

The problem must respond to the concrete needs of the company or the professional community:

- It must address a real and current issue, such as improving data quality, automating analytical processes, or implementing solutions enabling better decision-making using data.
- It can include goals such as reducing costs, increasing model accuracy, or identifying new insights from existing data.

This issue must be included in the technical or strategic priorities of the company, while contributing to the progression of practices and know-how in the field of data analysis.

This requirement to clarify a problem is not a purely academic constraint. Indeed, the formalization of this problem considerably helps the participant in his reflection and makes his approach more effective. It is also a privileged means of validating, through this final-year project, the ability to integrate the approaches, methods, concepts and tools developed during the PGE.

It is therefore appropriate, of course, to rule out a final-year project that, in narrative mode, would only be a presentation of the host company, its activity, its operation, and the work carried out during the mission. In the same way, we will rule out a final-year project that would only be a pure theoretical reflection. We will also prohibit a final-year project that would only be a purely empirical work (only data collection) evacuating any recourse to approaches, methods, concepts and tools seen during the PGE. We would similarly rule out a final-year project that would artificially attempt to combine a set of approaches, methods, concepts and tools seen during the PGE, without prior analysis, without link to an empiric, without prior sorting, without link to the problem addressed. Thus, the final-year project will have to take on a completely different and broader dimension with regard to a report that the host company could request as part of the in-company mission.

2 - A methodology

This section specifies an analysis framework as well as the methods used for data collection, preparation, and processing. The methodological approach must be described precisely and justified, taking into account the specificities of the data and the objectives of the project.

PFE Guide for PGE

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- **Analysis framework** : Clearly define the technical and analytical objectives (e.g., building a predictive model, optimizing a classification algorithm, or developing an interactive dashboard). Identify the variables of interest, the relationships to explore, and the performance indicators.
- **Data collection methods** : Describe how the data was collected (extraction from existing databases, web scraping , IoT sensors, APIs, etc.), specifying their source, nature (structured/unstructured), volume, and quality.
- **Data processing and analysis** : Justify the techniques chosen for data preparation (cleaning, normalization, treatment of missing values) and the tools or algorithms used for their analysis (statistical models, machine learning , data visualization).

Complete transparency in the process is essential. This includes:

- Description of the steps taken to ensure data integrity and reproducibility of results.
- Documentation of all methodological choices, such as the selection of algorithms or hyperparameters, with their justifications.

Limitations of the methodology should be identified, mentioned, and explained. This may include:

- Biases or inadequacies in the available data.
- The simplifying assumptions adopted to conduct the analysis.
- Technical or time constraints that influenced the scope of the project.

This structured framework helps to guarantee the scientific rigor and operational usefulness of the results produced as part of the final studies project.

For confidentiality reasons, it is entirely acceptable that the name of the company and/or operation(s) used for the PFE be hidden.

This work of clarifying the methodology is essential. It reveals the perspective that the participant is able to take in relation to a problem being addressed. It also allows the reader to validate the conclusions that the participant highlights and to identify any potential limitations.

3 - Results and recommendations

The results of a final-year project cannot be limited to a description of a situation. They must be analyzed and **provide concrete elements to answer the initial problem.**

The results arise from the methodology implemented with regard to a given problem. It is appropriate to find a suitable way of presenting these results. Thus, if interviews were conducted with different interlocutors, the interview reports can be annexed but the synthetic analysis of the data collected and the presentation of the results of this analysis must appear in the body of the thesis.

Based on the results of the analysis, a number of recommendations should be developed. Care should be taken to ensure that these recommendations are prioritized. This prioritization can be done on the basis of the nature of the recommendations (technical, commercial, organizational, etc.), their intensity (from the simplest to implement to the most complex), their urgency (from the short to the medium or long term, etc.).

Finally, it may be interesting to question the possible extension of the conclusions to situations other than the one investigated.

4 – Other essential parts

Any final year project obviously includes an “ executive summary » setting out the problem, the analysis methodology and the conclusions reached.

The thesis also includes a **brief presentation** of the host company (activities, turnover, workforce, products, locations, organization chart, etc.) and its industry, essentially aimed at providing an understanding of the context in which the participant operates, at what level of the organization he or she is attached, and what the position of the company is in its competitive environment.

A glossary clarifying technical terms or abbreviations used in the body of the text is welcome.

A table of contents is essential.

Finally, a bibliography is mandatory: it must include scientific and/or academic references and not just professional references.

CHAPTER III – Formal aspects of the thesis

"Form is substance rising to the surface" - Victor Hugo

❖ The length of the PFE thesis

The body of the thesis will be approximately **50 pages long** (excluding annexes, cover pages, acknowledgements pages, executive summary, summaries and bibliography).

❖ The cover page

The first page **must be** printed on aivancity letterhead. A copy of this letterhead will be sent to you by email. The following information must appear on this first page:

Aivancity Logo

Host company

Title of the final project

Final year project

Programme Grande Ecole

Year of submission

Participant: First and Last Name

Supervising Professor: Title First and Last Name

The cover page should be followed by a page for : 1) Abstract of the work 2) Acknowledgement and Declaration 3) Table of Contents mentioning section number 4) List of tables and figures. Following these four elements the body of the report should start on a separate page with the Chapter 1 of the report which could be the Introduction section. Report templates will be provided if required.

❖ Bibliographic references, plagiarism and use of generative AIs

You absolutely must rely on an appropriate bibliography. It allows you to expand on certain points covered in the PGE, and possibly initiate new reflections.

References are introduced in the body of the thesis. They appear in parentheses with the author's name and the date of publication, whether it is a book or an article. For example: (Dumoulin, Flipo et al., 1998).

At the end of the thesis, in a section entitled "Bibliographic references", there is a complete list of documents cited in their extensive form. For example: Prunier R., Mourel P. (1996), *l'Ingénierie Financière dans la relation Banque-Entreprise*, Economica, page 73 to 75.

Great caution must be exercised with quotations, whether they are excerpts from articles, books or interviews.

For any information obtained on the Internet, a specific source must be identified, both in the body of

the text in parentheses and in the bibliography.

Any excerpt from a book, article, website or other public or private source that would be included in the text of the final project, no matter how short, has a huge probability of being detected by the plagiarism detection software. If such an excerpt is not referenced as indicated above, both in the body of the text and in the bibliography, plagiarism will be proven and the final project grade will definitely be 0 (zero), with absolute impossibility of having a second chance, therefore absolute impossibility of ever graduating.

In addition, the use of generative AI (such as ChatGPT or any other similar tools) is permitted within the framework of the final year project, but only in compliance with the following rules:

1. Full transparency: Any use of generative AI must be explicitly mentioned, with a clear description of how it was used (e.g. to generate an idea, write a paragraph, analyze data, create a visualization, etc.).
2. Systematic reference: Each extract or content resulting from a generative AI must be referenced, preferably by indicating the tool used, the generation date, and a capture or a copy of the prompt used (attached if necessary).
3. Fair Use: Generative AI should be viewed as an assistive tool, not a substitute for personal intellectual work. The project should primarily reflect the student's own skills, ideas, and efforts.
4. Ethics: The use of generative AI must respect the values of integrity and responsibility. For example, it is strictly forbidden to use them to create fictitious or misleading content, or to circumvent methodological requirements.
5. Limitation of abuse: Overuse of generative AI (for example, to write the entire report or produce all of the analyses) will be considered as an attack on the spirit and educational objectives of the project.

Important: Work or parts generated by generative AIs can be completely misunderstood if they lack meaning or do not reflect in-depth reflection, supported by the student's critical perspective and personal expertise. The use of generative AI does not replace the need to demonstrate solid thinking, analysis and synthesis skills, which are the pillars of successful academic work. Meaningless content, even well formulated, will be penalized.

Finally, the school reserves the right to use generative AI detection tools to analyze submitted work. Excessive or unjustified use of these tools will be sanctioned, and may lead to the non-validation of the dissertation. These controls and rules aim to encourage responsible and ethical use of digital tools while ensuring the authenticity and quality of academic work. Any breach of these principles will be treated as a serious breach of the plagiarism rules, with the same severe consequences.

❖ Others

The entire document must be page numbered except the title page and the next four elements (abstract, acknowledgement and declaration, table of contents, list of tables and figures).

In the case of annexes, a summary of the annexes must be included. Each annex must be identified by a title.

CHAPTER IV – Relations between the student and the supervising professor

The organization of the relationship between student and supervising professor is to be defined jointly between the two. However, **some key steps are essential. The supervising professor will be particularly vigilant in ensuring that these commitments are respected.**

First step (February) :

The student must present and have the following elements validated by his supervisor:

- **Research question** : Three proposed formulations should be submitted.
- **Research problem** : A 2 to 3 page document (version 1).
- **Bibliographic references** : A total of 10 references, including at least 5 scientific articles.
- **Summary presentation of the empirical part envisaged** (1 to 2 pages):
 - o Describe the project: data, methods, expected results, type(s) of deliverables.OR
 - o Describe the empirical study: Why this study? What empirical methods? (type of data collected, analysis methods, hypotheses).

Second stage (March) :

The student must submit an intermediate document (**Report phase 1**) including:

- **Problem** : An updated version.
- **Detailed plan** : An organized and precise structure of the thesis.
- **Theoretical part** : A synthesis of the literature including the bibliographical references selected and enriched.
- **Implementation phase** :
 - o Types of data to be collected.
 - o Detailed presentation of the AI/Data techniques considered.
 - o Description of the tools or solutions that will be implemented.

This document must be validated by the supervisor before the student can continue.

Third stage (April) : Students must report progress, identify problems. Assignments of specific professors for projects.

Fourth stage (June) : Results and analysis work to be finalized. Beginning of compilation of PFE report

Fifth stage (July) : Verification of PFE report by supervising professor and PGE 2nd cycle coordinator. Discussions for feedback and inputs to report.

Final stage (September): Final PFE presentation

Interaction with the supervisor :

Once the previous steps have been validated, the student can move forward with finalizing their final project and writing their dissertation.

- The student is **solely responsible** for seeking advice or validation from his or her supervisor.
- Under no circumstances is the supervisor supposed to contact the student to ensure follow-up.

Submission and defense :

The final document must be submitted on the precise date indicated by the program management, as well as on the day scheduled for the defense. Any late submission or failure to comply with the schedule will be penalized in accordance with the regulations in force.

CHAPTER V – The Supervising Professor

The supervising professor plays a methodological supervisory role. He is therefore both a source of expertise for the participant and the guarantor of the application of the rules regarding the completion of the final project and the defense. Throughout the monitoring, the professor's assessment focuses on the conditions of validity and the limits of the work carried out, that is to say on all or part of the following points:

- Is the problem relevant? Well formalized? Well specified? Understandable?
- Are the methods used valid in relation to the current situation? Could other methods have been used, why were they not?
- Are the results obtained debatable? In view of these results, would it be relevant to supplement the methods used? How? By what other methods? Would this lead to refining the results, putting them into perspective or calling them into question?
- Is the student's thinking relevant? How reliable is it? What are its limits?
- Are the results obtained understandable? Can they be compared with other more general results or results obtained in other circumstances? Does this comparison or confrontation lead to giving another meaning to the results obtained?
- What generalization can be made about the approach followed and the methods used?
- Is the structure of the thesis (see next point) admissible?

CHAPTER VI – Thesis defense

1 - The defense: right to support and organization of the defense

The defense of the final project is done in the presence of the Supervising Professor and a qualified assessor (often the Director of the program, a permanent professor at aivancity) and if the student wishes, the internship supervisor in the host company or, failing that, a representative of the host company who does not participate in the deliberation.

It is the Supervising Professor **who alone decides to grant the student the right to defend** his or her final project. This defense is only granted from the moment when the final project is definitively acceptable in light of the elements recalled above.

The supervising professor gives his agreement for the participant to support his PFE.

2 - During the defense

The student will have to present, in 20 minutes, his approach and his results. This presentation must not be a rereading of the final project. The form of the presentation will be important in the evaluation.

A 20-minute slot will then be devoted to questions and a debate with the jury members. Finally, a ten-minute debriefing will be given to the student by the jury members.

At the end of this defense, the jury will award a mark for the PFE and for the defense.

5 - Confidentiality of the PFE

The final project that the PGE participant is required to write as part of the PGE may be of a confidential nature that the company wishes to keep and not have disclosed. In this case, and at the request of the company, a confidentiality document, co-signed by the company, the student and aivancity, could be drawn up.

7 - Don't forget!

- 📌 **Your mission must be directly linked to the teachings developed within the framework of the PGE .** Although this point is obvious *at first glance*, the temptation can sometimes be great to let yourself be influenced by the name, reputation and activity of a company, while the PFE that it will be possible to get from it is very far from the areas covered during the PGE. Know how to resist this temptation!
- 📌 **Your mission must truly allow you to develop a final-year project.** As such, it is important to ensure that the idea of a final-year project is fully integrated by the company that hosts you. This is why you must very quickly determine the problem on which you are going to work. The ideal is to get the company to grant you a little time to start writing this thesis during the last month. You can try to negotiate it, but of course, under no circumstances impose it.

- ❓ **A mission may pose confidentiality issues.** You must respect this confidentiality. However, you must ensure that this constraint will not represent too great a limitation for the writing of your PFE. It is important to raise this problem from the beginning of your mission. If the company does not want your work to be consulted by people other than the members of the jury, it is advisable to indicate this on the cover page and in the foreword of your thesis. A confidentiality letter can be signed by the head of the PGE .
- ❓ **The success of your project also depends on your supervision.** A supervising professor is assigned to you based on the subject of your project. His mission is certainly to evaluate you at the end of the final project defense, but he also represents fundamental *methodological support* during the internship. It is also important to benefit from a "follower" in the company, the work-study tutor or the internship supervisor, who can guide you and respond to your "daily" problems. It is, in fact, often difficult to be left to your own devices for this type of mission.

The PGE final studies project can be a real professional springboard, that it must be your business card, the demonstration of your ability to summarize, your technical skills and your desire to integrate.